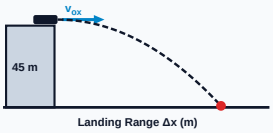
CAST Science Investigation: Projectile Motion Data Analysis

UNIT 2 WRAP-UP • DATA TABLES, TRENDS & SCIENTIFIC ARGUMENTS (FORM A)

Name: _____ **Per:** _____ **Date:** _____

EXPERIMENTAL SETUP: PHYSICS LAB LAUNCHER TESTS

Gravity: g = 10 m/s²



- **Investigation 1 (Horizontal Speed Test):** A spring launcher on a cliff fires balls horizontally so every trial has the exact same flight time (***t*** = **3 s**) while **Horizontal Speed (v_{ox})** increases.
- **Investigation 2 (Launch Angle Test):** On flat ground at constant speed v_o = **20 m/s**, students test five **Launch Angles (*θ*)** from **15°** to **75°**.

✏️ **Scan Card Instructions (Right):** Write & bubble your **6-digit Student ID** clearly inside the box on the right. **Leave the SCORE bubbles blank** for your teacher to bubble & scan!

CAST Projectile Data Analysis **MAX SCORE: 10**

Unit 2 Wrap-Up Investigation (Form A)

Bubble your 6-digit Student ID. Leave SCORE blank for teacher.

STUDENT ID

☐	☐	☐	☐	☐	☐
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

SCORE

☐	☐	/ 10
0	0	
1	1	
2	2	
3	3	
4	4	
5	5	
6	6	
7	7	
8	8	
9	9	

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INVESTIGATION 1: EFFECT OF HORIZONTAL LAUNCH SPEED (V_{OX}) ON LANDING RANGE (ΔX)

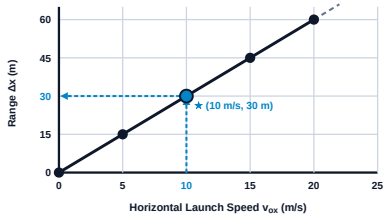
Flight Time: t = 3 s

Table 1: Horizontal Cliff Launch Data

Trial	Horizontal Speed v_{ox} (m/s)	Flight Time t (s)	Landing Range Δx (m)
1	0	3	0
2	5	3	15
3 ★	10	3	30
4	15	3	45
5	20	3	60

💡 **How to Read Point ★ on Graph 1:**
1. Find **10 m/s** on the bottom axis (v_{ox}).
2. Follow the dashed arrow UP to the line, then LEFT to read **30 m**.

Graph 1: Landing Range (Δx) vs. Horizontal Speed (v_{ox})



INVESTIGATION 2: EFFECT OF LAUNCH ANGLE (Θ) ON FLIGHT TIME & LANDING RANGE

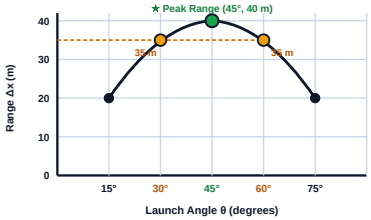
Launch Speed: v_o = 20 m/s

Table 2: Angled Ground Launch Data (v_o = 20 m/s)

Launch Angle θ (°)	Horizontal Speed v_{ox} (m/s)	Hang Time t_{hang} (s)	Landing Range Δx (m)
15°	19.3	1.0	20
30° ●	17.3	2.0	35
45° ★	14.1	2.8	40 (Max)
60° ●	10.0	3.5	35
75°	5.2	3.9	20

🔍 **Notice the Pattern in Table 2:**
- As angle increases from **15° to 75°**, Hang Time (t_{hang}) always **increases**.
- Complementary angles that add to **90°** (like **30° & 60°**) land at the **exact same distance (35 m)**!

Graph 2: Landing Range (Δx) vs. Launch Angle (θ)



PART 1: READING THE DATA TABLES & IDENTIFYING PATTERNS (2 POINTS)

Look at Table 1 & Graph 1

(1) READ DATA Look at **Table 1 and Graph 1**:

- **a) Worked Example:** When Horizontal Speed (v_{ox}) is **10 m/s**, the Landing Range (Δx) is **30 m**.
- **b) Your Turn:** When Horizontal Speed (v_{ox}) is **15 m/s**, the Landing Range (Δx) is _____ **m**. When v_{ox} is **20 m/s**, the Landing Range (Δx) is _____ **m**.

(2) TREND **Circle ONE word in each bracket** to describe the pattern in **Graph 1**:
As Horizontal Speed (v_{ox}) increases, Landing Range (Δx) **[increases / decreases / stays the same]** at a constant rate, creating a **[straight linear line / curved arch]**.

PART 2: CALCULATING RATE OF CHANGE & MAKING A PREDICTION (2 POINTS)

Stacked Math Practice

(3A) WORKED PATTERN: RATE OF GRAPH 1

Every +1 m/s of horizontal speed adds +3 m of distance because the flight time is 3 s:

$t = \Delta x \div v_{ox}$

$t = 30\ m \div 10\ m/s = 3\ s$

(3B) YOUR TURN: PREDICT TRIAL 6 (V_{OX} = 25 M/S)

If students launch a ball at v_{ox} = 25 m/s with the same flight time (t = 3 s), how far will it land?

$\Delta x = v_{ox} \times t$

$\Delta x =$ _____ m/s × 3 s = _____ m

PART 3: CHECKING LAUNCH ANGLE PATTERNS IN INVESTIGATION 2 (2 POINTS)

Look at Table 2 & Graph 2

(4A) PEAK ANGLE According to **Table 2 and Graph 2**, which launch angle (θ) produces the **farthest Landing Range (40 m)**?
Circle ONE: [**15°** / **30°** / **45°** / **60°** / **75°**]

(4B) MATCHING PAIR Look at **15°** in Table 2 (Landing Range = **20 m**). Which **other launch angle** lands at the exact same **20 m** distance, and which of the two stays in the air longer?

- Matching Angle: _____ °
- Stays in the air longer (higher t_{hang}): _____ ° (Hang Time = _____ **s**)

PART 4: CAST SCIENTIFIC ARGUMENT — CLAIM, EVIDENCE & REASONING (4 POINTS)

Complete the Sentence Starters

❓ **Student Misconception Question:** A student claims: *"Shooting a projectile at a steeper angle (like 60° or 75°) ALWAYS makes it land farther away because it stays in the air longer."* Use **Table 2 and Graph 2** to explain why this student's claim is **incorrect**.

☐ **ZONE A — CLAIM (1 pt): State whether the student is correct or incorrect and state the true pattern.**

The student's claim is **[circle one: correct / incorrect]**. In Investigation 2, landing range increases from **15°** up to a maximum at _____ °, and then **[increases / decreases]** at steeper angles like **60°** and **75°**.

☐ **ZONE B — EVIDENCE (2 pts): Fill in two data points from Table 2 (include numbers and units!).**

- **Data Point 1:** At a launch angle of **45°**, the hang time is **2.8 s** and the landing range reaches a maximum of _____ **m**.
- **Data Point 2:** When the angle is increased to **60°** (or **75°**), the hang time increases to _____ **s**, but the landing range drops down to _____ **m**.

☐ **ZONE C — REASONING (1 pt): Connect the evidence to the physics formula $\Delta x = v_{ox} \times t_{hang}$.**

Because horizontal landing range equals **Horizontal Speed (v_{ox}) × Hang Time (t_{hang})**, shooting at a very steep angle gives the ball more _____ time in the air, but leaves it with too little horizontal _____ (v_{ox} drops to only **10.0 m/s** at 60° and **5.2 m/s** at 75°).